Luxembourg and Timor-Leste Case Studies Recreated With the Scalable Distance Pipeline

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1 Purpose

This note restates the Luxembourg school-access and Timor-Leste health-facility case studies using the reengineered `scalable_general_distances_per_country` pipeline. The goal is to describe the same analytical narrative as the original paper, but in terms of the new modular contracts: shared cache manifests, country source descriptors, source and target tables, interchangeable routing strategies, sparse distance matrices, and optimization-ready coverage neighborhoods.

The companion notebook `luxembourg_api_illustration.ipynb` contains executable miniature versions of the cases. Those examples are intentionally small enough to run on a fresh development machine, while the final notebook cells show the country scale API and CLI calls that use the shared cache.

2 Pipeline Architecture Used by Both Cases

Both case studies use the same sequence of self-contained modules.

1. **Country source resolution.** `CountryDataSources` resolves the Geofabrik PBF filename, WorldPop filename, download URLs, and deterministic cache paths before any data are downloaded. Geofabrik publishes country and regional OpenStreetMap extracts in `.osm.pbf` form, and its technical notes describe the daily extract process based on OSM tooling such as Osmium [1, 2].

2. **Shared cache and manifest.** Downloaded artefacts are stored under the shared cache used by the original implementation. The cache layout and manifest make it explicit which files are downloaded, generated, or reused.

3. **Data extraction.** The production runner extracts a drivable network from the OSM PBF, converts WorldPop raster cells to target points, and extracts OSM facilities matching the requested amenity values. OSM tags such as `amenity=school`, `amenity=hospital`, and `amenity=clinic` are part of the community-maintained map-feature vocabulary [3].

4. **Routing strategy.** The router is selected by name. `NetworkXRouter` is the portable default and uses NetworkX shortest-path routines for weighted graphs [6]. `PandanaRouter` is imported only when selected; Pandana is designed for network accessibility and fast shortest-path queries using contraction hierarchies [7], so deployment environments without a compatible Pandana and NumPy stack can still use the package.

5. **Distance matrix output.** The same source and target tables feed the router, and the matrix writer can produce combined, split, or both output layouts. The split layout supports

large-country runs by keeping source-target distance chunks lightweight. The same strategy boundary can support future engines such as R5, Conveyal's multimodal accessibility routing engine [8].

6. **Downstream analysis.** Descriptive coverage and prescriptive maximum-covering models consume the sparse matrix without depending on the implementation details of the routing backend.

3 Data Sources, Data Types, and Persistence

The reengineered pipeline separates source-data identity from in-memory data representation. The country descriptor records the expected filenames and URLs; the cache records whether an artefact was downloaded or generated; and the pipeline modules exchange explicit source, target, network, and matrix tables.

Object	Typical source format	Pipeline role
Road network	`.osm.pbf` from Geofabrik	Parsed into node and edge tables with stable node IDs, longitude, latitude, and edge length in metres.
Facilities	OSM tags in the same `.osm.pbf`	Filtered into source rows with `source_id`, `source_type`, coordinates, and amenity attributes.
Population	WorldPop GeoTIFF raster	Converted into target rows with `target_id`, coordinates, and represented population. GeoTIFF is the OGC standard for georeferenced TIFF imagery [5].
Distance matrix	Generated table	Stored as combined or split source-target rows with source ID, target ID, network distance, and snapping diagnostics.

Table 1: Main data objects and their pipeline roles.

The table contract is intentionally compatible with either pandas or Polars. Both libraries expose two-dimensional DataFrame objects for tabular data, while the pipeline keeps the public contract at the column-schema level rather than at the implementation level [9, 10]. Persistence is likewise a boundary concern. CSV is useful for transparent row-oriented exchange [11]; XLSX is useful when analysts need Excel-compatible workbooks [12]; and Parquet is useful for larger analytical tables because it is a columnar file format with an explicit specification [13]. The pipeline therefore treats persistence as an interchangeable writer choice, not as part of the routing or geocoding logic.

Case	Paper settings	Paper-scale reported result
Luxembourg schools	OSM `amenity=school`; no population aggregation; retained matrix threshold of 1,000 m; context map saved.	394 OSM school features; 102,061 population targets representing 478,010 people; 266,225 people within 1,000 m of an OSM school.
Timor-Leste health	Default health amenities plus generated 5 km candidate sites; aggregate factor 8; retained matrix threshold of 10 km; maximum covering with $p = 20$.	28,162 population targets; 860 sources; 100,997 retained distance rows; Gurobi and HiGHS both cover 757,053 represented people at 10 km.

Table 2: Paper case definitions restated as scalable-pipeline inputs.

4 Case Definitions

5 Luxembourg School Access

The Luxembourg case is descriptive. OSM schools are treated as existing source locations and WorldPop cells are treated as population targets. WorldPop population layers are gridded raster products, so the pipeline converts raster cell values into weighted point targets before routing [4]. The reengineered pipeline resolves the country data as:

```
CountryDataSources(
    country_slug="luxembourg",
    iso3="LUX",
    base_dir=shared_cache / "luxembourg_data",
    worldpop_filename="lux_ppp_2020.tif",
)
```

The production configuration is:

```
ProductionRunConfig(
    sources=luxembourg_sources,
    output_dir=shared_cache / "luxembourg_data" / "outputs",
    run_tag="luxembourg_schools_networkx",
    amenity_values=("school",),
    router="networkx",
    matrix_output_mode="both",
)
```

For each target i, the analysis asks whether at least one school j is within a road-network threshold R:

$$\text{covered}_i(R) = \begin{cases} 1 & \text{if } \min_{j \in J_{\text{school}}} d_{ij} \le R, \\ 0 & \text{otherwise.} \end{cases}$$

The population covered at threshold R is then

$$\sum_i w_i \, \text{covered}_i(R),$$

where w_i is the represented population of target i. A target is counted only once per threshold even if several schools are reachable. For school ranking, the same matrix can assign each covered target to its nearest school.

The corresponding CLI call is:

```
py -m scalable_distances.cli run ^
  --country-slug luxembourg ^
  --iso3 LUX ^
  --base-dir C:\local\Download_Depot\luxembourg_data ^
  --output-dir C:\local\Download_Depot\luxembourg_data\outputs ^
  --run-tag luxembourg_schools_networkx ^
  --amenity school ^
  --router networkx ^
  --matrix-output-mode both
```

6 Timor-Leste Health Facility Coverage

The Timor-Leste case is prescriptive. It builds a sparse matrix between population targets and health-related source locations, then chooses a limited number of sources to maximize covered population. The model follows the maximal covering location problem introduced by Church and ReVelle [14]. The reengineered pipeline resolves the country data as:

```
CountryDataSources(
    country_slug="east-timor",
    iso3="TLS",
    base_dir=shared_cache / "east-timor_data",
    worldpop_filename="tls_ppp_2020.tif",
)
```

The health amenity run is configured as:

```
ProductionRunConfig(
    sources=timor_leste_sources,
    output_dir=shared_cache / "east-timor_data" / "outputs",
    run_tag="timor_leste_health_networkx",
    amenity_values=(
        "hospital", "clinic", "doctors", "dentist", "pharmacy",
        "health_post", "nursing_home", "social_facility",
    ),
    router="networkx",
    aggregate_factor=8,
    matrix_output_mode="both",
)
```

The paper case also includes generated 5 km candidate sites. In the new design, candidate generation belongs to the same source-table contract as OSM amenities: candidate points can be added as an additional source layer before snapping and routing, while the router and matrix writer remain unchanged.

Let $N(i)$ be the set of source locations within 10 km of target i, based on the sparse routing matrix. The maximum-covering model is:

$$\max \sum_i w_i z_i$$

$$z_i \leq \sum_{j \in N(i)} y_j \quad \forall i, \qquad \sum_j y_j \leq p, \qquad z_i, y_j \in \{0, 1\}.$$

Here $y_j = 1$ if source j is selected and $z_i = 1$ if population target i is covered. In the paper-scale case, $p = 20$, the coverage radius is 10 km, and both Gurobi and HiGHS reach the same covered population.

The corresponding CLI scaffold is:

```
py -m scalable_distances.cli run ^
  --country-slug east-timor ^
  --iso3 TLS ^
  --base-dir C:\local\Download_Depot\east-timor_data ^
  --output-dir C:\local\Download_Depot\east-timor_data\outputs ^
  --run-tag timor_leste_health_networkx ^
  --amenity hospital --amenity clinic --amenity doctors ^
  --amenity dentist --amenity pharmacy --amenity health_post ^
  --amenity nursing_home --amenity social_facility ^
  --router networkx ^
  --aggregate-factor 8 ^
  --matrix-output-mode both
```

7 Reproducibility Checks

The new pipeline makes the following checks explicit for both cases:

- source-data filenames and URLs are resolved before download;
- downloads are cached in the shared cache and reused on repeated runs;
- source and target tables expose stable IDs, coordinates, geometry, and attributes;
- route computation records router name, network size, source count, target count, and matrix row count;
- split and combined matrix outputs can be compared from the same run;
- Pandana is not imported unless the Pandana strategy is requested.

These checks allow the original implementation and the scalable implementation to be compared at the level of cache contents, road-network inputs, coordinate handling, snapping, origin-destination pairs, distance calculations, and final summary statistics.

8 Interpretation

The core scientific narrative is unchanged: the Luxembourg case measures the population already close to schools, while the Timor-Leste case uses the same distance-matrix representation to support facility-location decisions. The engineering narrative changes substantially. Instead of one monolithic script, the reengineered version exposes lightweight data contracts, lazy routing strategies, explicit persistence choices, and cache metadata that can be reused by notebooks, command-line runs, tests, and future geocoding or optimization modules.

References

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